

IIR filter design

Impulse invariance technique

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Design of IIR filters from Analog filters

Analog filter system function can be described

1) Using filter coefficients

$$H_a(s) = \frac{B(s)}{A(s)} = \frac{\sum_{k=0}^M \beta_k s^k}{\sum_{k=0}^N \alpha_k s^k} = \frac{\beta_0 + \beta_1 s + \beta_2 s^2 + \dots + \beta_M s^M}{\alpha_0 + \alpha_1 s + \alpha_2 s^2 + \dots + \alpha_N s^N}$$

$\{\alpha_k\}$ and $\{\beta_k\}$ - analog filter coefficients

2) Using impulse response

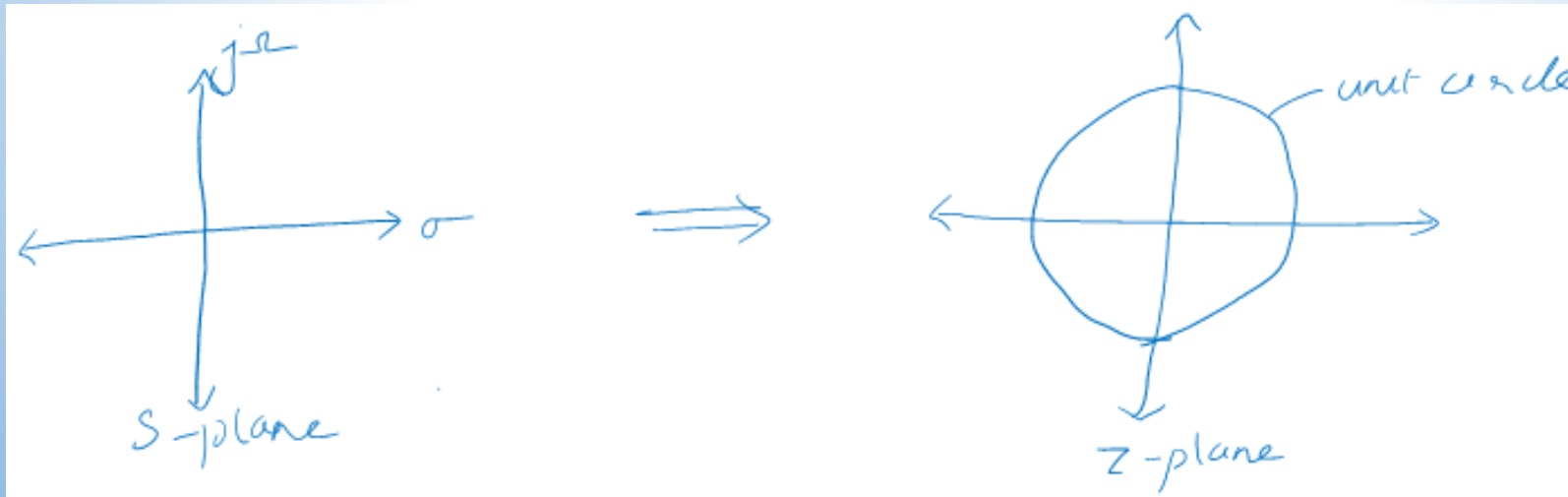
$$H_a(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt$$

3) Using LCCDE

$$\sum_{k=0}^N \alpha_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M \beta_k \frac{d^k x(t)}{dt^k}$$

Properties for effective conversion from Analog to Digital (mapping from s-plane to z-plane)

- Direct relationship between the frequency variable in the s-domain and z-domain
 - $j\Omega$ axis in s-plane should map into the unit circle in z-plane
- LHP of s-plane should map into the inside of the unit circle in z-plane.
 - A stable analog filter can be converted to a stable digital filter
 - Analog LTI system is stable if all its poles lie in the left half of s-plane



Linear phase filter must have a system function that satisfies the condition:

$$H(z) = \pm z^{-N} H(z^{-1})$$

Symmetric and antisymmetric response

$$h(n) = \pm h(N-n)$$

$\downarrow z^T$

$$H(z) = \sum_{n=0}^N h(n) z^{-n} = h(0) + h(1)z^{-1} + \dots + h_N z^{-N}$$

$$H(z^{-1}) = \sum_{k=0}^N h(k) z^k = h(0) + h(1)z + \dots + h(N)z^N$$

$$z^{-N} H(z^{-1}) = h(0) z^{-N} + h(1) z^{-N+1} + \dots + h(N-1) z^{-1} + h(N) z^0$$

If condition $H(z) = \pm z^{-N} H(z^{-1})$ is met,
 $h(0) = h(N)$, $h(1) = h(N-1)$, ...

This condition implies the roots of polynomial $H(z)$ are identical to the roots of polynomial $H(z^{-1})$

- Roots of $H(z)$ must occur in reciprocal pairs.

If z_1 is a root of $H(z)$, $\frac{1}{z_1}$ is also a root.

For every pole inside a unit circle, there will be a mirror image pole outside the unit circle.

Therefore the system is unstable.

A causal and stable IIR system cannot have linear phase.

Design of IIR filters:

- Two steps
- First design an analog filter – Butterworth, Chebyshev or Elliptic
- Convert this analog filter to digital using
 - Impulse invariance technique
 - Bilinear transformation
 - Matched Z – transform

Impulse invariance transformation:

We design an IIR filter having unit-sample response

$h(n)$ = sampled version of IR of analog filter

$$h(n) \equiv h(nT) \quad , n = 0, 1, 2, \dots$$

↑

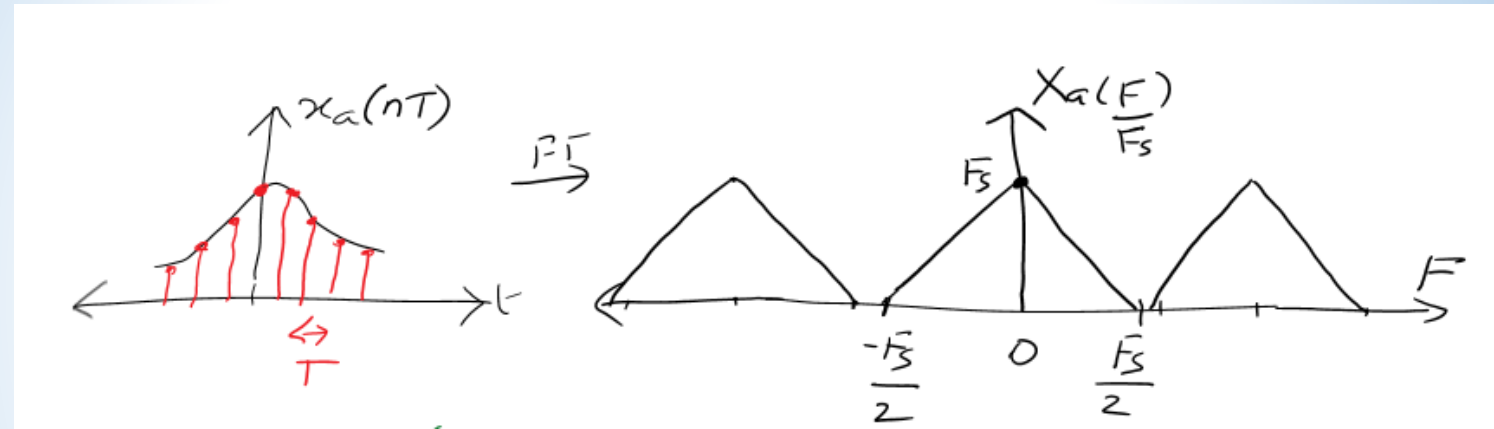
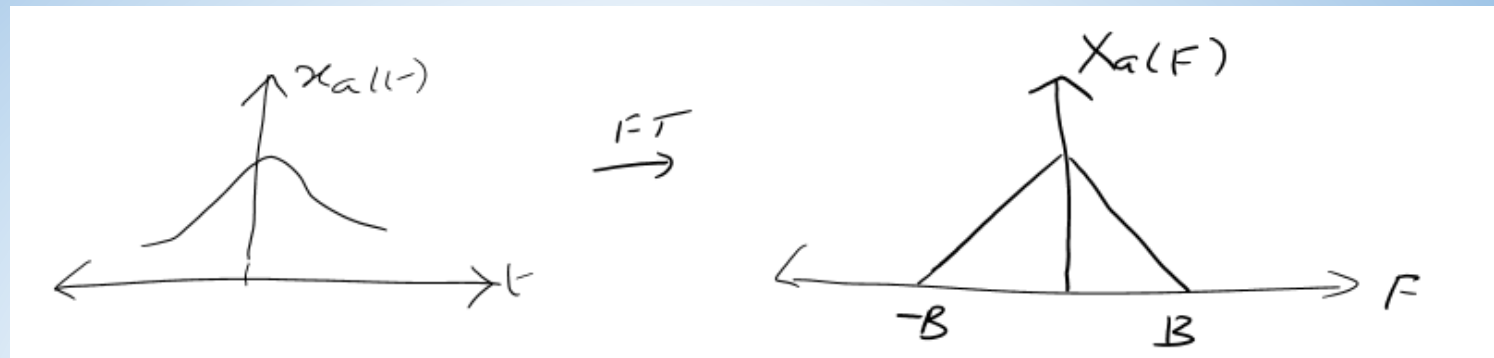
T = sampling interval

When a CT signal having a spectrum $X_a(f)$ is sampled at a rate $F_s = \frac{1}{T}$ samples per sec,

the spectrum of the sampled signal $x_a(nT)$

is the periodic repetition of the scaled

spectrum $F_s X_a(f)$, with period F_s .



$$\begin{aligned}
 \therefore X(f) &= F_s \sum_{k=-\infty}^{\infty} X_a(f - k F_s) F_s, \quad f = \frac{F}{F_s} \\
 &= F_s \left[\cdots X_a\left(\frac{F}{F_s} - 1\right) F_s + X_a\left(\frac{F}{F_s}\right) F_s + X_a\left(\frac{F}{F_s} + 1\right) F_s + \cdots \right] \\
 &= F_s \left[\cdots X_a(F - F_s) + X_a(F) + X_a(F + F_s) + \cdots \right]
 \end{aligned}$$

Aliasing occurs if the sampling rate
 $F_s < 2F_{\max}$ of $X_a(f)$
 $T > 2T_{\max}$

As sampling interval T increases,
the copies of the spectrum move closer
Spectral overlap $\rightarrow$ aliasing

With respect to sampling the IR of analog filter
with frequency response $H_a(f)$,
the digital filter with unit-sample response
 $h(n) \equiv h_a(nT)$ has the frequency response

$$H(f) = F_s \sum_{k=-\infty}^{\infty} H_a(f - k) F_s$$

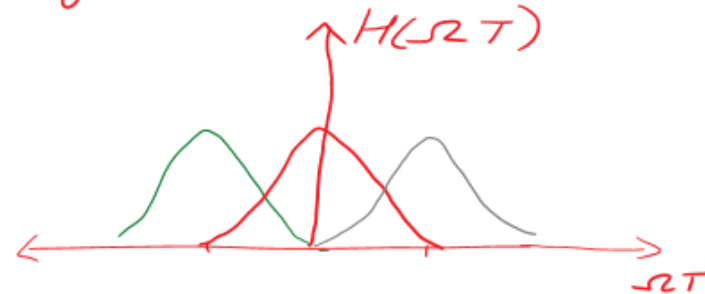
$$H(f) = F_s \sum_{k=-\infty}^{\infty} H_a[(f-k) F_s]$$

$$H(\omega) = F_s \sum_{k=-\infty}^{\infty} H_a(\omega - 2\pi k) F_s]$$

ω is periodic with 2π

$$\begin{aligned} \downarrow \Omega T \\ H(\Omega T) &= \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(\Omega T - 2\pi k) \frac{1}{T} \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(\Omega - \frac{2\pi k}{T}) \end{aligned}$$

Frequency response of analog filter:



Effects of aliasing can be minimized if the sampling interval T is selected sufficiently small.

Mapping of points between z-plane & s-plane implied by the sampling process:

Relation between z-transform of $h(n)$ and Laplace transform of $h_a(t)$:

$$\begin{aligned} H_a(s) &= \int_0^{\infty} h_a(t) e^{-st} dt \\ H_a(s) \Big|_{\text{sampled at } nT} &= \int_0^{\infty} \sum_{n=0}^{\infty} h(n) \delta(t-nT) e^{-st} dt \\ &= \sum_{n=0}^{\infty} h(n) \int_0^{\infty} \delta(t-nT) e^{-st} dt \\ &= \sum_{n=0}^{\infty} h(n) e^{-snT} \quad \left[\because \mathcal{L}(\delta(t-a)) = e^{-sa} \right] \\ &= H(z) \Big|_{z=e^{sT}} \quad \left[H(z) = \sum_{n=0}^{\infty} h(n) z^{-n} \right] \end{aligned}$$

Frequency response of digital filter with unit sample resp. $h_a(nT)$:

$$\begin{aligned} H(\Omega T) &= \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(s - j\frac{2\pi k}{T}) \\ \Rightarrow H(z) \Big|_{z=e^{sT}} &= \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a(s - j\frac{2\pi k}{T}) \quad \Leftarrow \text{Relation} \end{aligned}$$

Mapping points from s-plane to z-plane:

$$z = e^{sT}$$

$$re^{j\omega} = e^{(\sigma + j\Omega)T} = e^{\sigma T} \cdot e^{j\Omega T}$$

$$\therefore r = e^{\sigma T}$$

$$\omega = \Omega T$$

$\sigma < 0, 0 < r < 1 \Rightarrow$ LHP in s-plane is mapped inside unit- $\odot$ in z-plane

$\sigma > 0, r > 1 \Rightarrow$ RHP in s-plane is mapped outside the unit $\odot$ in z-plane

$\sigma = 0, r = 1 \Rightarrow$ $j\Omega$ axis is mapped onto the unit $\odot$ in z-plane.

Mapping of $j\Omega$ axis onto the unit circle is not one-to-one

Mapping $\omega = \Omega T$,
Analog freq interval maps into freq interval in digital

$$-\frac{\pi}{T} \leq \Omega \leq \frac{\pi}{T}$$

$$-\pi \leq \omega \leq \pi$$

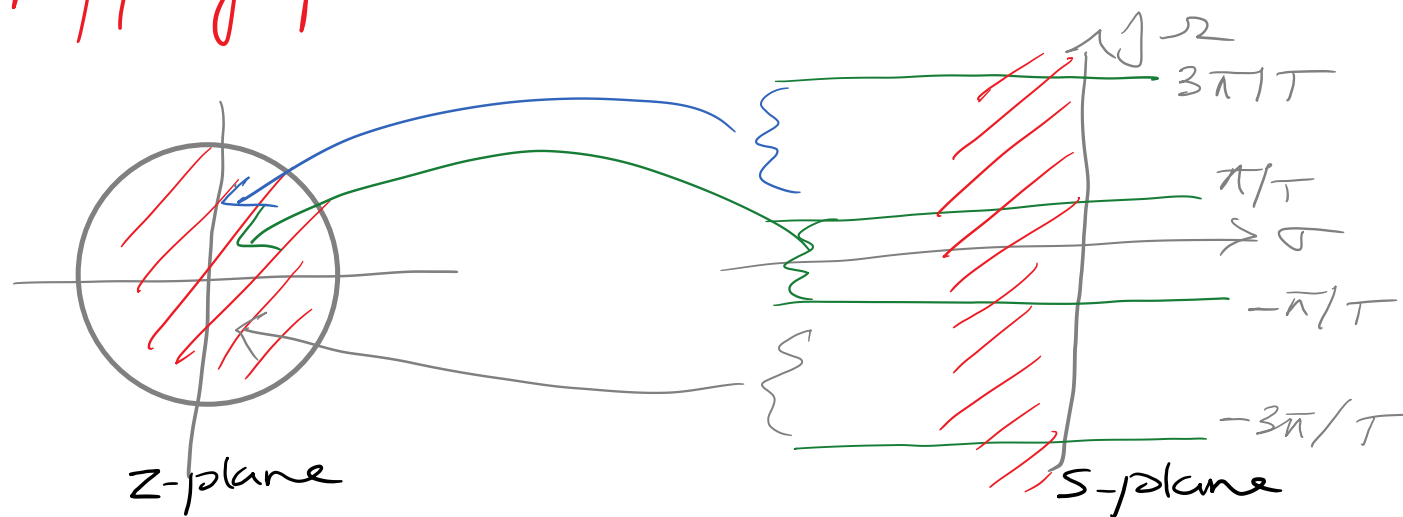
$$\frac{\pi}{T} \leq \Omega \leq \frac{3\pi}{T}$$

$$-\pi \leq \omega \leq \pi$$

$$(2k+1)\frac{\pi}{T} \leq \Omega \leq (2k+1)\frac{\pi}{T}$$

$$-\pi \leq \omega \leq \pi, \quad k = \text{integers}$$

$\therefore$ Mapping from Ω to ω is many-to-one



Let us express the system function of analog filter in partial fraction form.

Assuming distinct poles of analog filter:

$$H_a(s) = \sum_{k=1}^N \frac{C_k}{s - p_k} \quad \leftarrow \begin{array}{l} \text{coefficients in partial fraction} \\ \text{distinct poles} \end{array}$$

$$h_a(t) = \sum_{k=1}^N C_k e^{p_k t}, \quad t \geq 0 \quad \left[\mathcal{L}^{-1} \left[\frac{1}{s-a} \right] = e^{at} \right]$$

Sampling $h_a(t)$ periodically at $t = nT$,

$$h(n) = h_a(nT)$$

$$= \sum_{k=1}^N C_k e^{p_k nT}$$

System function of resulting digital IIR filter,

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n}, \quad \text{substitute } h(n)$$

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \sum_{k=1}^N c_k e^{p_k n T} \cdot z^{-n}$$

$$= \sum_{k=1}^N c_k \sum_{n=0}^{\infty} (e^{p_k T} z^{-1})^n$$

$$\frac{1}{1 - e^{p_k T} \cdot z^{-1}}$$

Geometric series exp

$$\sum_{i=0}^{\infty} a_i r^i = \frac{a_1}{1-r}$$

$$H(z) = \sum_{k=1}^N \frac{c_k}{1 - e^{p_k T} z^{-1}}$$

We observe that the digital filter has poles at

$$z_k = e^{p_k T}, \quad k = 1, 2, \dots, N$$

Impulse invariance transformation

$$H(z) = \sum_{k=1}^N \frac{c_k}{1 - e^{p_k T} z^{-1}}$$

We observe that the digital filter has poles at

$$z_k = e^{p_k T}, \quad k = 1, 2, \dots, N$$

This equation holds for IIR filter having distinct poles.

Due to presence of aliasing, impulse invariance is suitable for LPIF and BPF only.

Convert the analog filter with system function

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter by means of the impulse invariance method.

Solution. We note that the analog filter has a zero at $s = -0.1$ and a pair of complex-conjugate poles at

$$p_k = -0.1 \pm j3$$

from the partial-fraction expansion of $H_a(s)$

$$H(s) = \frac{\frac{1}{2}}{s + 0.1 - j3} + \frac{\frac{1}{2}}{s + 0.1 + j3}$$

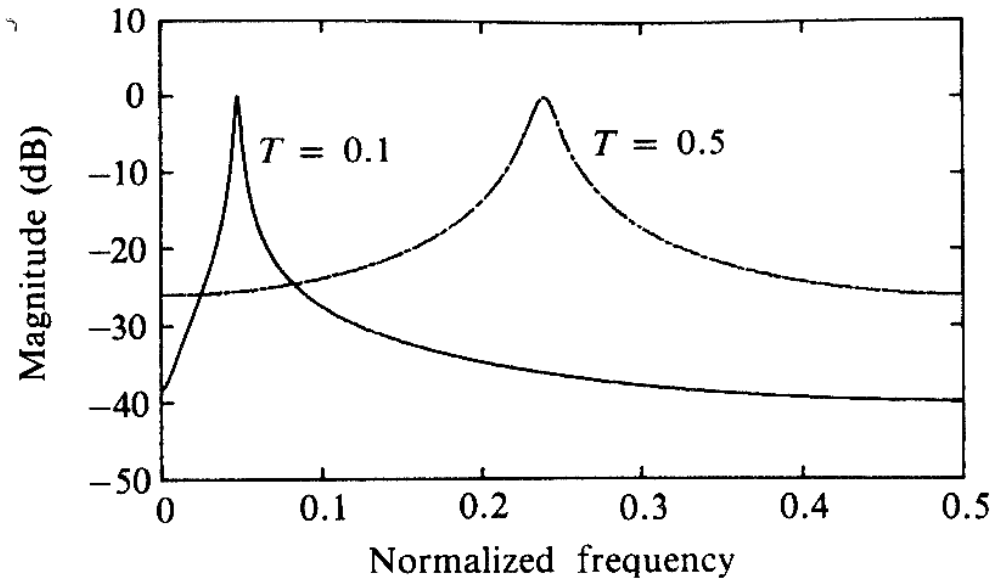
$$H(z) = \frac{\frac{1}{2}}{1 - e^{-0.1T} e^{j3T} z^{-1}} + \frac{\frac{1}{2}}{1 - e^{-0.1T} e^{-j3T} z^{-1}}$$

Remember this

$$\frac{s+a}{(s+a)^2 + b^2} \Rightarrow \frac{1 - e^{-aT} \cos bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$$

$$\frac{b}{(s+a)^2 + b^2} \Rightarrow \frac{e^{-aT} \sin bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$$

$$H(z) = \frac{1 - (e^{-0.1T} \cos 3T)z^{-1}}{1 - (2e^{-0.1T} \cos 3T)z^{-1} + e^{-0.2T}z^{-2}}$$



We note that aliasing is significantly more prevalent when $T = 0.5$ than when $T = 0.1$. Also, note the shift of the resonant frequency as T changes.

*Thank
you*

